

E2E WORKFLOW & DATA FLOW

Terra Brew Coffee Traceability Platform

Complete system architecture, role-based workflow maps, and end-to-end data flow guide for the multi-tenant coffee supply chain traceability platform. Covering seven entity types, eight user roles, and thirty-five functional modules across six global tenants.

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Terra Brew Engineering Team

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Chapter 1: Platform Architecture Overview

1.1 Multi-Entity Architecture

The Terra Brew Coffee Traceability Platform is built on a multi-entity architecture that mirrors the real-world coffee supply chain. Seven distinct entity types form the backbone of the system, each representing a critical node in the journey of coffee from seed to cup. These entity types are: **Producer**, **Aggregator**, **Exporter**, **Importer**, **Buyer**, **Certification Body**, and **Laboratory**. Each entity type has its own data model, workflow, permission scheme, and module visibility configuration, ensuring that organizations only see and interact with the parts of the platform relevant to their role in the supply chain.

The Producer entity represents coffee farmers and farming cooperatives who cultivate coffee plants, manage land, and harvest cherries. Aggregators, such as cooperatives and mill operators, collect raw cherries from multiple producers, process them through washing, drying, and milling, and create exportable lots. Exporters manage the commercial side of moving processed coffee across borders, handling RFQs, smart contracts, export documentation, and regulatory compliance including EUDR. Importers receive shipments at destination ports, manage customs clearance, and distribute to domestic buyers. Buyers browse the marketplace, submit RFQs, sign contracts, and track shipments to their final destinations. Certification Bodies perform farm audits, compliance assessments, and issue certificates for organic, fair trade, and other standards. Laboratories conduct quality inspections, cupping tests, and generate QC reports that validate the quality grade of each lot.

The interaction between these entities is not merely sequential; it forms a rich network of data exchanges. Producers send harvest data to Aggregators, who combine it into lots with quality data from Laboratories. Exporters aggregate lot data into shipment manifests with EUDR compliance documentation. Importers verify shipment data and pass traceability records to Buyers. Certification Bodies provide independent verification at multiple points in this chain. This architecture ensures that every piece of data has a clear provenance and every handoff is recorded immutably on the blockchain layer.

1.2 Role-Based Access Control

The platform implements a fine-grained role-based access control (RBAC) system with eight distinct roles that can be assigned across all seven entity types. This matrix-based approach ensures that users only access features relevant to their organizational role, preventing data leaks and maintaining operational security. The eight roles are: **tenant_admin**, **operations_manager**, **field_officer**, **quality_controller**, **trader**, **finance_manager**, **buyer**, and **viewer**. Each role carries a specific set of permissions that determine which modules are accessible and what actions (create, read, update, delete, approve) can be performed within each module.

The following matrix illustrates which roles are applicable to each entity type. A checkmark indicates that the role is valid and commonly assigned within that entity type, while a dash indicates

that the role is not applicable or extremely rare for that entity type. This matrix serves as the foundation for user provisioning and permission management.

Role	Producer	Aggregator	Exporter	Importer	Buyer	Cert Body	Lab
tenant_admin	Yes	Yes	Yes	Yes	Yes	Yes	Yes
operations_manager	No	Yes	Yes	Yes	No	No	No
field_officer	Yes	No	No	No	No	No	No
quality_controller	No	Yes	No	No	No	Yes	Yes
trader	No	Yes	Yes	Yes	No	No	No
finance_manager	Yes	Yes	Yes	Yes	No	No	No
buyer	No	No	Yes	No	Yes	No	No
viewer	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Table 1.2: Role-Entity Applicability Matrix

1.3 Multi-Tenant Data Isolation

The Terra Brew platform is a multi-tenant SaaS application where each tenant represents a distinct organization within the coffee supply chain. Data isolation is enforced at the database level through tenant scoping, ensuring that no organization can access or modify data belonging to another tenant. The platform currently supports six tenants, each configured with a specific entity type, geographic region, local currency, and language preferences.

Metrang Coffee (Vietnam, VND) operates as a Producer, managing smallholder farmers in the Central Highlands region. Cooxupe (Brazil, BRL) is an Aggregator, one of the largest coffee cooperatives in the world, processing cherries from thousands of member farmers. Yirgacheffe Union (Ethiopia, ETB) is a Producer cooperative famous for its heirloom varieties. Othaya Cooperative (Kenya, KES) is another Producer entity in the Nyeri highlands. Euro Coffee Imports (Netherlands, EUR) operates as an Exporter, facilitating the movement of African and South American coffees into European markets. SGS Inspection (Switzerland, CHF) is a Certification Body providing independent third-party auditing.

Tenant isolation is implemented through a combination of row-level security policies and application-level filtering. Every API request includes a tenant identifier extracted from the authenticated user session, and all database queries automatically append tenant-scoped WHERE clauses. Cross-tenant data sharing occurs only through explicitly defined handoff points, such as

when an Aggregator receives harvest data from a Producer tenant, or when an Exporter creates a shipment manifest that references lots from an Aggregator tenant. These handoffs use a separate integration layer with strict validation and audit logging. The platform supports five languages (Vietnamese, English, Portuguese, Amharic, and Swahili), with each tenant configured to display the UI in the preferred language of its user base.

Tenant	Entity Type	Country	Currency	Language
Metrang Coffee	Producer	Vietnam	VND	vi / en
Cooxupe	Aggregator	Brazil	BRL	pt / en
Yirgacheffe Union	Producer	Ethiopia	ETB	am / en
Othaya Cooperative	Producer	Kenya	KES	sw / en
Euro Coffee Imports	Exporter	Netherlands	EUR	en
SGS Inspection	Certification Body	Switzerland	CHF	en

Table 1.3: Multi-Tenant Configuration

1.4 Module Visibility Matrix

The platform comprises 35 modules organized into 7 functional groups. Each module has a visibility level for each entity type: **Full** (complete read-write access), **View** (read-only access), or **Hidden** (module is not visible in the sidebar). This matrix determines the navigation structure that each user sees after logging in, ensuring that organizations only interact with modules relevant to their business operations. The following table provides the complete module visibility mapping across all seven entity types.

Module	Producer	Aggregator	Exporter	Importer	Buyer	Cert Body	Lab
Dashboard	Full	Full	Full	Full	Full	Full	Full
Analytics	Full	Full	Full	Full	View	View	View
Farmers	Full	Hidden	Hidden	Hidden	Hidden	View	Hidden
Farm Lands	Full	Hidden	Hidden	Hidden	Hidden	View	Hidden
Cultivations	Full	Hidden	Hidden	Hidden	Hidden	Hidden	Hidden
Nurseries	Full	Hidden	Hidden	Hidden	Hidden	Hidden	Hidden
Land Prep	Full	Hidden	Hidden	Hidden	Hidden	Hidden	Hidden
Crop Monitor	Full	Hidden	Hidden	Hidden	Hidden	Hidden	Hidden

Fertilizer	Full	Hidden	Hidden	Hidden	Hidden	Hidden	Hidden
Pest Disease	Full	Hidden	Hidden	Hidden	Hidden	Hidden	Hidden
Harvest Trace	Full	View	Hidden	Hidden	View	Hidden	Hidden
Procurement	Full	Full	Hidden	Hidden	View	Hidden	Hidden
Processing	Hidden	Full	Hidden	Hidden	View	Hidden	Hidden
Coffee Insp	View	View	Hidden	Hidden	View	Full	Full
QC Verify	Hidden	Full	View	Full	View	Hidden	Full
EUDR	Full	Full	Full	Full	View	Full	Hidden
Cert Assess	Full	Full	View	View	View	Full	Hidden

Table 1.4a: Module Visibility Matrix (Part 1 of 2)

Module	Producer	Aggregator	Exporter	Importer	Buyer	Cert Body	Lab
Deforestation	Full	Full	View	Hidden	View	Full	Hidden
Marketplace	Full	Full	Full	Full	Full	Hidden	Hidden
RFQ	View	Full	Full	Full	Full	Hidden	Hidden
Inspections	View	Full	Full	Full	View	Full	Full
Product Monitor	View	Full	Full	Full	View	Hidden	Hidden
Smart Contracts	Hidden	Full	Full	Full	Full	Hidden	Hidden
Trading Desk	Hidden	Full	Full	Full	Full	Hidden	Hidden
Shipments	Hidden	Full	Full	Full	View	Hidden	Hidden
Logistics	Hidden	Full	Full	Full	View	Hidden	Hidden
Export Docs	Hidden	Hidden	Full	Hidden	View	Hidden	Hidden
Buyers	Hidden	Full	Full	Full	Full	Hidden	Hidden
Trace Journey	Full	Full	Full	Full	Full	View	Hidden
Billing	Full	Full	Full	Full	View	Full	Full
Users	Full	Full	Full	Full	Hidden	Full	Full
IoT Sensors	Full	Full	Hidden	Hidden	Hidden	Hidden	Hidden
Blockchain	Full	Full	Full	Full	View	Hidden	Hidden
API Settings	Full	Full	Full	Full	Hidden	Hidden	Hidden

Chapter 2: E2E Coffee Supply Chain Data Flow

2.1 The Coffee Journey: Seed to Cup

The journey of coffee from seed to cup traverses multiple organizations, geographies, and regulatory regimes. Understanding this journey is essential to grasping why the Terra Brew platform structures its data flows the way it does. The coffee supply chain begins at the nursery, where selected varieties are germinated and nurtured into seedlings. These seedlings are transplanted to prepared farm land, where they grow for three to five years before producing their first viable harvest. During the cultivation phase, farmers manage fertilization schedules, monitor for pest and disease outbreaks, and track environmental conditions that affect crop health.

When cherries reach optimal ripeness, they are selectively picked by hand (for specialty coffee) or strip-picked (for commodity grades). The harvest trace module records picking dates, quantities, varietal information, and the specific plot of land from which each batch originated. This data becomes the first link in the traceability chain. Cherries are then transported to a processing facility, typically operated by an Aggregator, where they undergo one of several processing methods: washed (wet), natural (dry), honey (semi-washed), or experimental anaerobic fermentation. Each processing method creates distinct flavor profiles and is recorded with precise parameters including fermentation times, temperatures, and drying durations.

After processing, the green coffee is milled to remove parchment, sorted by size and density, and graded by quality. Laboratories perform cupping tests following SCA protocols, assigning scores that determine market value. The graded lots are listed on the marketplace where Exporters and Importers negotiate prices through RFQs and smart contracts. Exporters prepare shipping documentation, including bills of lading, phytosanitary certificates, and EUDR Due Diligence Statements. The coffee travels by sea freight to the destination port, where the Importer clears customs and arranges inland transport. Finally, the coffee reaches the Buyer, who may be a roaster, retailer, or consumer brand, completing the journey from seed to cup with a complete digital traceability record accessible via QR code.

2.2 Data Entry Points

Data entry in the Terra Brew platform is distributed across roles and entity types, with each user contributing specific data at defined stages of the supply chain. This distributed data entry model ensures that information is captured at the source by the person most qualified to provide it, reducing transcription errors and ensuring data freshness. The Field Officer is the primary data entry point for the Producer entity, enrolling farmers, mapping farm boundaries using GPS, recording cultivation activities, and logging harvest data. The Operations Manager at an Aggregator enters processing batch data, lot creation details, and inventory movements. Quality Controllers at

Laboratories and Certification Bodies enter inspection results, cupping scores, and compliance assessment outcomes. Traders manage RFQs, negotiate prices, and execute smart contracts. Finance Managers handle billing, payment reconciliation, and cost tracking. Buyers submit RFQs and confirm receipt of shipments.

Each data entry point has built-in validation rules. For example, harvest quantities must fall within expected ranges based on the farm size and historical yields. Cupping scores must follow the 0-100 SCA scale. GPS coordinates must fall within recognized coffee-growing regions. Smart contract values must match the negotiated terms from the RFQ process. These validations prevent common data quality issues and ensure that downstream analytics and compliance reports are built on reliable data. The platform also supports IoT sensor integration, allowing automated data capture for temperature, humidity, and soil moisture readings, reducing the manual data entry burden on Field Officers.

Supply Chain Stage	Data Entered	Primary Role	Entity Type
Nursery	Seedling batches, varietal, planting dates	Field Officer	Producer
Land Preparation	Plot boundaries, soil analysis, altitude	Field Officer	Producer
Cultivation	Fertilizer schedules, pest treatments, pruning	Field Officer	Producer
Crop Monitoring	Growth stages, flowering, cherry development	Field Officer	Producer
Harvest	Cherry quantities, picking dates, plot ID	Field Officer	Producer
Procurement	Delivery weights, cherry prices, farmer payments	Ops Manager	Aggregator
Processing	Method, fermentation time, drying duration	Ops Manager	Aggregator
Quality Inspection	Cupping scores, defect counts, moisture %	QC Inspector	Lab / Cert Body
Lot Creation	Lot composition, grade, warehouse location	Ops Manager	Aggregator
Marketplace Listing	Lot details, price, minimum order qty	Trader	Aggregator / Exporter
RFQ	Quantity, quality spec, delivery terms	Trader / Buyer	Exporter / Importer
Smart Contract	Contract terms, price, shipment schedule	Trader	Exporter / Importer
Export Documentation	B/L, phytosanitary cert, EUDR DDS	Ops Manager	Exporter
Shipment Tracking	Vessel, ETA, container number	Trader	Exporter / Importer
Customs Clearance	Duty paid, customs release date	Ops Manager	Importer

Delivery Confirmation	Received qty, quality check at destination	Buyer	Buyer
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Table 2.2: Data Entry Points by Supply Chain Stage

2.3 Verification and Quality Gates

Quality gates are critical control points in the data flow where information is verified, validated, and either approved or rejected. These gates prevent erroneous or fraudulent data from propagating through the supply chain, maintaining the integrity of the traceability record. The first quality gate occurs at procurement, where the Aggregator verifies that the delivered cherry weight matches the farmer-reported harvest quantity within an acceptable tolerance (typically plus or minus 5 percent). Significant discrepancies trigger an investigation workflow. The second quality gate is at processing, where the Aggregator confirms that the processing method and parameters align with the lot classification. The third and most stringent quality gate is the laboratory inspection, where certified Q-graders perform blind cupping tests and assign scores that determine the lot grade and market value.

EUDR compliance represents another critical quality gate. Before a shipment can be exported, the Deforestation Due Diligence Statement must be generated, requiring satellite imagery verification that the farm land was not deforested after December 31, 2020. The platform integrates with satellite monitoring services to automate this verification. Smart contract execution is also gated, requiring both parties to digitally sign the contract terms, which are then recorded on the blockchain for immutability. At the final delivery stage, the Buyer confirms receipt and can verify the complete traceability chain from the original farm to the delivered shipment, providing the ultimate quality gate in the consumer-facing traceability workflow.

2.4 Cross-Entity Data Flow

Data flows between entities in the Terra Brew platform follow a structured handoff protocol that ensures data integrity and traceability across organizational boundaries. The primary data flow follows the physical supply chain: Producer to Aggregator to Exporter to Importer to Buyer. However, there are also lateral data flows to Certification Bodies and Laboratories that operate independently of the primary chain. When a Producer records harvest data, it is stored within the Producer tenant. When the Aggregator receives the cherries, a procurement record is created that references the original harvest data through a cross-tenant reference. This reference is not a direct database join but rather a secure API call that retrieves a hash-verified copy of the data, ensuring that the Aggregator sees the Producer data but cannot modify it.

The Exporter receives lot data from the Aggregator, including quality grades, lot compositions, and origin details. This data is used to create marketplace listings and respond to RFQs. When a smart contract is signed, the lot data, contract terms, and compliance documents are bundled into a shipment manifest. The Importer receives this manifest and uses it for customs clearance. The

Buyer receives a subset of the shipment data through the Trace Journey module, which presents a consumer-friendly view of the coffee journey. Throughout this chain, the blockchain layer records hashes of each data handoff, creating an immutable audit trail. If any party disputes a data point, the blockchain record can be used to verify the original data at the time of handoff. This cross-entity data flow architecture ensures that the traceability chain remains unbroken from the first data entry point to the final consumer query.

Chapter 3: Role-by-Role Workflow Maps

3.1 Tenant Administrator

The Tenant Administrator is the most privileged role in the Terra Brew platform, responsible for the complete configuration and management of the tenant environment. This role has unrestricted access to all modules available to the tenant entity type and can perform all CRUD operations. The primary responsibilities of a Tenant Administrator include user management (creating, modifying, and deactivating user accounts, assigning roles, and managing permissions), tenant configuration (setting currency, language, timezone, and branding preferences), module configuration (enabling or disabling specific modules and adjusting visibility settings), billing management (viewing invoices, managing payment methods, and tracking subscription usage), and API settings (configuring webhooks, managing API keys, and setting up third-party integrations).

Upon first login, the Tenant Administrator is greeted by the Dashboard, which provides an executive overview of key metrics, recent activity, and system health indicators. From here, the administrator navigates to the Users module to provision accounts for team members, assigning each user the appropriate role based on their job function. The Billing module displays the current subscription plan, usage statistics, and invoice history. The API Settings module allows the administrator to generate API keys for integration with ERP systems, accounting software, or custom applications. The Tenant Administrator also has the authority to configure data validation rules, set approval workflows for quality gates, and define notification preferences for the entire organization. This role is critical during the initial platform setup phase and ongoing system administration.

3.2 Operations Manager

The Operations Manager is the day-to-day orchestrator of supply chain activities within the Aggregator and Exporter entity types. This role focuses on the efficient flow of coffee from receipt to shipment, ensuring that processing schedules are met, quality standards are maintained, and inventory is accurately tracked. The Operations Manager oversees the procurement process, recording cherry deliveries from farmers, weighing and grading incoming lots, and initiating farmer payments through the Finance Manager. During processing, the Operations Manager monitors fermentation tanks, drying beds, and milling equipment, recording batch parameters and processing outcomes.

The Operations Manager creates lots from processed coffee, assigning quality grades based on laboratory results and preparing lots for marketplace listing. For Exporter entities, this role also manages export documentation, including phytosanitary certificates, certificates of origin, and EUDR Due Diligence Statements. The Operations Manager coordinates with the Trader role to ensure that lots are listed at competitive prices and that shipment schedules align with contract delivery terms. Key modules for this role include Procurement, Processing, QC Verify, Shipments, Logistics, and Export Docs. The Dashboard provides operational KPIs such as processing throughput, inventory levels, and pending shipment statuses, enabling data-driven decision-making for resource allocation and process optimization.

3.3 Field Officer

The Field Officer is the frontline data collector for the Producer entity type, responsible for enrolling farmers, mapping farm boundaries, recording cultivation activities, and logging harvest data. This role is unique to the Producer entity and is critical for establishing the foundational data layer of the traceability chain. Field Officers typically operate in remote rural areas with limited internet connectivity, so the platform supports offline data capture with automatic synchronization when connectivity is restored. The primary modules for the Field Officer are Farmers, Farm Lands, Cultivations, Nurseries, Land Prep, Crop Monitor, Fertilizer, Pest Disease, and Harvest Trace.

The farmer enrollment workflow begins with the Field Officer visiting a farm, collecting the farmer personal details, and capturing GPS coordinates of the farm boundary using the mobile application. The farm land mapping feature uses polygon drawing tools to define the exact boundaries of each plot, which is essential for EUDR compliance verification. During the cultivation phase, the Field Officer records planting dates, varietal selection, fertilizer applications, and pest or disease treatments. As the crop matures, the Field Officer monitors growth stages, flowering periods, and cherry development. At harvest time, the Field Officer records the quantity of cherries picked from each plot, the picking method (selective or strip), and the date of harvest. All this data forms the provenance record that follows the coffee through the entire supply chain, providing the traceability foundation that end consumers can verify through QR codes on the final product packaging.

3.4 Quality Controller

The Quality Controller role is shared across three entity types: Aggregator, Certification Body, and Laboratory. At each entity, the focus differs but the core responsibility remains the same: ensuring that quality standards are met and documented. For Aggregators, the Quality Controller performs in-house inspections of processed coffee, checking moisture content, defect counts, bean size distribution, and conducting preliminary cupping assessments. For Certification Bodies, the Quality Controller conducts formal farm audits, assessing compliance with organic, fair trade, Rainforest Alliance, or other certification standards. For Laboratories, the Quality Controller performs definitive quality testing using standardized protocols, generating official QC reports.

The quality control workflow begins with sample receipt, where the Quality Controller logs the incoming sample, verifies its provenance through the lot reference number, and assigns it a unique

sample identifier. Physical inspection follows, including visual assessment for defects, moisture measurement using calibrated instruments, and bean size grading using standard sieves. Cupping is performed following SCA (Specialty Coffee Association) protocols: the sample is roasted to a light profile, rested for 8-24 hours, ground, and evaluated for fragrance, aroma, flavor, aftertaste, acidity, body, balance, uniformity, clean cup, sweetness, and overall impression. Each attribute is scored on a 0-10 scale, with the total cupping score determining the quality grade. The Quality Controller records all findings in the platform, generating a QC report that becomes part of the permanent traceability record. For Certification Bodies, the workflow extends to scheduling and conducting farm audits, reviewing farmer documentation, and issuing or renewing certificates based on compliance scores.

3.5 Trader

The Trader role is the commercial engine of the Terra Brew platform, responsible for managing the marketplace, responding to RFQs, negotiating prices, and executing smart contracts. This role is available in the Aggregator, Exporter, and Importer entity types. For Aggregators, the Trader lists processed lots on the marketplace, setting prices and minimum order quantities. For Exporters, the Trader responds to buyer RFQs, negotiates contract terms, and manages the smart contract lifecycle from creation through execution. For Importers, the Trader sources coffee from multiple origins, submitting RFQs to Exporters and managing the procurement pipeline.

The trading workflow begins when a Trader reviews available inventory in the Marketplace module and creates a listing with lot details, quality grade, price per unit, and available quantity. When a buyer submits an RFQ, the Trader receives a notification and can accept, counter-offer, or decline. Price negotiation occurs through the RFQ messaging system, with each counter-offer recorded for audit purposes. Once terms are agreed, the Trader initiates a Smart Contract, which captures the commercial terms (price, quantity, quality spec, delivery date, payment terms) and presents them for digital signature by both parties. After signing, the contract is recorded on the blockchain, and the Trader moves to the Trading Desk to manage active positions and track market prices. The Trader also coordinates with the Operations Manager to ensure that shipment schedules align with contract delivery commitments, using the Shipments and Logistics modules to track the physical movement of coffee.

3.6 Finance Manager

The Finance Manager role oversees all financial operations within the platform, including billing, payments, procurement costs, and financial reporting. This role is available across Producer, Aggregator, Exporter, and Importer entity types. The Finance Manager has access to the Billing module, which displays the platform subscription details, usage metrics, and invoice history. For Producers, the Finance Manager tracks farmer payment schedules, ensuring that cherry delivery payments are calculated correctly based on weight, quality grade, and the prevailing market price. For Aggregators, the Finance Manager oversees procurement costs, processing expenses, and margin calculations for lot sales.

For Exporters and Importers, the Finance Manager handles the more complex financial flows associated with international trade, including letter of credit management, currency exchange calculations, and duty payments. The Billing module provides a comprehensive view of all financial transactions, with filters for date range, entity, and transaction type. The Finance Manager can generate financial reports that aggregate data across all modules, providing insights into profitability by origin, margin by quality grade, and cost breakdowns by supply chain stage. Payment reconciliation features allow the Finance Manager to match incoming payments against outstanding invoices, flagging discrepancies for review. The Users module is also accessible to the Finance Manager for reviewing user activity logs related to financial transactions, supporting audit compliance requirements.

3.7 Buyer

The Buyer role represents the demand side of the coffee supply chain, encompassing roasters, consumer brands, and retail organizations that purchase green coffee. This role is available in the Exporter and Buyer entity types. For the Exporter entity, the Buyer role represents internal purchasing agents who source coffee from origin markets. For the Buyer entity type, this role represents external organizations that purchase coffee through the marketplace. The Buyer workflow begins with browsing the Marketplace, where available lots are displayed with origin details, quality grades, cupping notes, and pricing information. The Buyer can filter lots by country, region, varietal, processing method, quality score, and price range.

When a suitable lot is identified, the Buyer submits an RFQ specifying the desired quantity, quality parameters, delivery terms, and target price. The RFQ is routed to the relevant Trader, who may accept, counter-offer, or decline. Upon agreement, the Buyer reviews and digitally signs the Smart Contract, which legally binds both parties to the transaction terms. After contract execution, the Buyer uses the Shipments module to track the physical movement of coffee from origin to destination, receiving real-time updates on vessel location, estimated arrival time, and customs clearance status. Upon delivery, the Buyer confirms receipt in the platform, recording the actual delivered quantity and conducting a quality verification. The Trace Journey module provides the Buyer with a complete provenance record, from the farm where the coffee was grown to the laboratory where it was graded, offering full supply chain transparency that can be shared with end consumers through QR code verification.

3.8 Viewer

The Viewer role provides read-only access to the Terra Brew platform, designed for stakeholders who need visibility into supply chain data without the ability to create, modify, or delete records. This role is available across all seven entity types and is commonly assigned to executives, board members, investors, regulatory inspectors, and external auditors who require access to reports and dashboards but should not be able to alter operational data. The Viewer can access the Dashboard, which presents key performance indicators and summary metrics relevant to the entity type. Viewers at Producer organizations can see farm-level statistics and harvest reports. Viewers at

Aggregator organizations can see processing throughput and inventory levels. Viewers at Exporter organizations can see shipment statuses and contract pipelines.

The Viewer role has access to the Analytics module, which provides advanced data visualization and reporting capabilities. Pre-built report templates include harvest yield analysis, quality trend reports, financial summaries, and compliance dashboards. Viewers can customize report parameters such as date range, geographic region, and quality grade, but cannot create or modify the underlying data. The Trace Journey module is also accessible in read-only mode, allowing Viewers to trace the complete provenance of any lot or shipment. The Viewer role is essential for maintaining organizational transparency and supporting data-driven governance without compromising data integrity. It also serves as a safe onboarding role for new users who need to familiarize themselves with the platform before being assigned operational permissions.

Chapter 4: Entity-Specific Workflows

4.1 Producer Workflow

The Producer workflow encompasses the entire agricultural lifecycle of coffee, from nursery management through harvest delivery. It begins with farmer enrollment, where the Field Officer visits the farm, collects the farmer personal information (name, national ID, contact details, GPS location), and creates a farmer profile in the platform. Next, the Field Officer maps the farm land using the mobile application GPS tools, drawing polygons that define the exact boundaries of each plot. This mapping is critical for EUDR compliance, as it enables satellite-based deforestation monitoring. The system records the plot area, altitude, slope aspect, and soil type.

During the cultivation phase, the Field Officer records all agricultural activities: nursery seedling planting, land preparation (clearing, terracing, shade management), planting of seedlings in the field, fertilizer applications (type, quantity, date), pest and disease treatments (chemical or organic), pruning activities, and crop monitoring observations (growth stage, flowering status, cherry development). Each activity is timestamped and geolocated, creating a detailed agronomic record. When cherries reach peak ripeness, the harvest is recorded in the Harvest Trace module, capturing the date, plot, quantity, varietal, and picking method. The harvested cherries are then delivered to the Aggregator, creating a procurement record that links the harvest data to the Aggregator processing batch. The Producer can also access the EUDR and Deforestation modules to view their compliance status and deforestation risk assessment. The Billing module shows farmer payment records and the Blockchain module displays the immutable hash chain for all Producer data.

4.2 Aggregator Workflow

The Aggregator workflow begins when cherries arrive at the processing facility. The Procurement module records each delivery, capturing the farmer ID, cherry weight, delivery date, and agreed price. A weighing ticket is generated and linked to the farmer profile. The Operations Manager oversees the processing workflow, which varies by processing method. For washed coffee, cherries

are depulped, fermented in tanks for 12-72 hours, washed to remove mucilage, and dried on raised beds or in mechanical dryers. For natural coffee, cherries are dried whole on patios or raised beds. Each processing batch is recorded in the Processing module with parameters including method, fermentation time, drying temperature, humidity, and duration.

After processing and drying to the target moisture content (typically 10-12 percent), the coffee is milled to remove parchment and sorted by bean size using standard sieves. The Quality Controller performs preliminary inspections, and samples are sent to Laboratories for official cupping and grading. Based on the laboratory results, the Operations Manager creates lots in the platform, combining processed coffee from multiple batches into marketable lots with consistent quality profiles. Each lot is assigned a unique identifier, quality grade, and warehouse location. The Trader then lists the lots on the Marketplace, setting prices and availability. When an RFQ is received, the Trader negotiates and, upon agreement, initiates a Smart Contract. The Aggregator also manages the EUDR compliance workflow, generating Due Diligence Statements for lots destined for EU markets. The Trace Journey module enables full backward tracing from any lot to the original farms and forward tracing to the Exporter who purchased the lot.

4.3 Exporter Workflow

The Exporter workflow centers on the commercial and logistical aspects of moving coffee from origin countries to destination markets. It begins with the Trader responding to RFQs from Importers and Buyers. The Exporter reviews available inventory (either own stock or lots sourced from Aggregators), evaluates the buyer requirements, and prepares a quotation. If the buyer accepts, a Smart Contract is created that captures the commercial terms: price, quantity, quality specification, delivery schedule, and payment terms. Both parties digitally sign the contract, which is then recorded on the blockchain for immutability.

After contract execution, the Operations Manager takes over to prepare the shipment. The Export Docs module is used to generate all required documentation: bill of lading, phytosanitary certificate, certificate of origin, packing list, and the EUDR Due Diligence Statement. The EUDR module is particularly critical for Exporters, as they are legally responsible for ensuring that coffee exported to the EU meets deforestation-free requirements. The Deforestation module provides satellite-based risk assessments for each farm contributing to the shipment. The Shipments module tracks the physical movement of containers from the warehouse to the port, onto the vessel, and through transit to the destination port. The Logistics module manages the coordination of inland transport, container stuffing, and port logistics. The Buyer role within the Exporter entity can also browse the marketplace to source additional lots from Aggregators to fulfill contract commitments.

4.4 Importer Workflow

The Importer workflow represents the receiving end of international coffee trade. The Trader at the Importer organization submits RFQs to Exporters, specifying the desired origin, quality, quantity, and delivery terms. Upon receiving quotations, the Trader evaluates options from multiple Exporters, considering price, quality, and reliability. After selecting a supplier, the Trader reviews

and signs the Smart Contract. The contract execution triggers the shipment tracking workflow. The Shipments module provides real-time visibility into the vessel location, estimated time of arrival, and container status.

When the shipment arrives at the destination port, the Operations Manager uses the Logistics module to coordinate customs clearance, arranging for the payment of import duties and obtaining customs release. The QC Verify module is used to record the results of quality checks performed on the received coffee, comparing the actual quality against the contract specifications. Discrepancies are flagged for resolution through the smart contract dispute mechanism. After clearance, the Operations Manager arranges inland transport to the final delivery location. The Billing module tracks all costs associated with the import process, including the purchase price, shipping costs, insurance, duties, and handling fees, providing a complete landed cost calculation. The Trace Journey module enables the Importer to trace any received shipment back to its origin farms, supporting the due diligence requirements of EUDR and other regulatory frameworks.

4.5 Buyer Workflow

The Buyer workflow focuses on sourcing, procurement, and traceability verification. The Buyer begins by browsing the Marketplace, which displays available lots from Aggregators and Exporters. The marketplace supports filtering by origin country, region, varietal, processing method, quality score (SCA grade), certification type, and price. Each listing includes detailed lot information, cupping notes, and a traceability summary. When the Buyer identifies a suitable lot, they submit an RFQ through the platform, specifying the desired quantity, quality parameters, delivery destination, and target price.

The RFQ process supports multiple rounds of negotiation, with each counter-offer recorded for transparency. Once terms are agreed, the Buyer reviews the Smart Contract, verifying all commercial and quality terms before digitally signing. After contract execution, the Buyer uses the Shipments module to track the delivery progress. Real-time updates include vessel departure, transit milestones, port arrival, customs clearance, and inland transport. Upon receiving the shipment, the Buyer confirms delivery in the platform, recording the actual received quantity and conducting a quality verification. The Trace Journey module provides the Buyer with a complete provenance record, from the farm where the coffee was grown through every processing, quality, and logistics step. This record can be shared with end consumers through QR codes printed on the final product packaging, enabling consumer-facing traceability that builds brand trust and supports sustainability claims.

4.6 Certification Body Workflow

The Certification Body workflow is centered on the assessment, auditing, and certification of farms, aggregators, and processing facilities against recognized standards such as organic, fair trade, Rainforest Alliance, UTZ, and 4C. The workflow begins with assessment scheduling, where the Certification Body assigns auditors to specific farms or facilities based on the certification program requirements and the client application timeline. The platform supports calendar management for

auditor scheduling, with integration to the Inspections module for tracking upcoming and completed audits.

During a farm audit, the Quality Controller (auditor) uses the platform to record observations across multiple compliance criteria. For organic certification, this includes verification of no prohibited substance use, buffer zone maintenance, and soil management practices. For fair trade certification, the auditor assesses labor practices, minimum price compliance, and community development fund contributions. Each criterion is scored, and the aggregate compliance score determines whether the certificate is issued, conditionally issued (with corrective actions required), or denied. The Cert Assess module provides the framework for recording all assessment data, including photographic evidence, GPS-verified locations, and auditor notes. When a certificate is issued, it is recorded in the platform with a validity period, and the system automatically generates renewal reminders. The EUDR module enables the Certification Body to provide third-party verification of deforestation-free status for farms seeking EU market access.

4.7 Laboratory Workflow

The Laboratory workflow follows a standardized quality testing protocol that produces authoritative quality reports used throughout the supply chain for pricing, classification, and regulatory compliance. The workflow begins with sample receipt, where the laboratory logs the incoming sample against a lot reference number, verifying the chain of custody and sample integrity. Each sample is assigned a unique identifier and stored under controlled conditions until testing begins. The Coffee Inspection module records the preliminary physical assessment, including visual defect count, moisture content measurement, water activity reading, and bean size distribution.

The QC Verify module captures the results of the formal quality testing process. Cupping is performed by certified Q-graders following SCA protocols, with each attribute scored individually on a 0-10 scale. The total cupping score determines the quality grade: Specialty (80+), Premium (75-79.99), Standard (70-74.99), or Below Standard (below 70). Additional tests may include chemical residue analysis for organic certification, mycotoxin screening for food safety compliance, and sensory analysis for flavor profiling. The laboratory generates a comprehensive QC report that is recorded in the platform and linked to the original lot. This report becomes part of the permanent traceability record and is referenced by Aggregators for lot grading, by Traders for pricing, and by Buyers for quality verification. The Billing module tracks laboratory service charges, and the Users module manages the certification credentials of Q-graders and laboratory technicians. The Inspections module is also used to manage the scheduling and tracking of on-site quality inspections at processing facilities.

Chapter 5: Data Entry and Reporting Guide

5.1 Data Entry Points by Module

Understanding which modules require data entry and which roles are responsible for entering data is essential for efficient platform operation. The following table maps each module to the roles that perform data entry within it, along with the types of data entered. Modules that are view-only or hidden for a given entity type are excluded from the data entry requirements. This guide helps organizations plan user provisioning and training, ensuring that every data entry point has a designated responsible user.

Module	Data Entry Role(s)	Primary Data Entered
Farmers	Field Officer	Farmer profile, contact, ID, location
Farm Lands	Field Officer	Plot boundaries (GPS), area, altitude, soil
Cultivations	Field Officer	Planting date, varietal, spacing, shade
Nurseries	Field Officer	Seedling batches, germination rate, planting
Land Prep	Field Officer	Clearing, terracing, shade management
Crop Monitor	Field Officer	Growth stage, flowering, cherry development
Fertilizer	Field Officer	Type, quantity, date, method of application
Pest Disease	Field Officer	Pest type, severity, treatment, date
Harvest Trace	Field Officer	Cherry weight, date, plot, picking method
Procurement	Ops Manager	Delivery weight, price, farmer ID, date
Processing	Ops Manager	Method, batch params, duration, output weight
Coffee Insp	QC Inspector	Defects, moisture, bean size, visual grade
QC Verify	QC Inspector	Cupping scores, quality grade, test results
EUDR	Tenant Admin / Ops	Farm polygons, DDS data, risk assessment
Cert Assess	QC Inspector	Compliance scores, audit findings, photos
Deforestation	Tenant Admin	Satellite data, risk flags, verification status
Marketplace	Trader	Lot listing, price, minimum order quantity
RFQ	Trader / Buyer	Quantity, quality spec, price, delivery terms
Smart Contracts	Trader	Contract terms, digital signature, status
Trading Desk	Trader	Position management, price alerts, market data
Shipments	Ops Manager / Trader	Container, vessel, ETA, tracking updates

Logistics	Ops Manager	Transport arrangements, customs, handling
Export Docs	Ops Manager	B/L, phytosanitary, cert of origin, DDS
Billing	Finance Manager	Payment records, cost allocation, invoicing
Users	Tenant Admin	User accounts, role assignments, permissions
IoT Sensors	Ops Manager	Sensor config, data thresholds, alerts
Blockchain	Tenant Admin	Chain verification, hash validation
API Settings	Tenant Admin	API keys, webhooks, integration config

Table 5.1: Data Entry Points by Module

5.2 Report Types by Role

Each role in the Terra Brew platform has access to specific report types that support their operational and strategic decision-making needs. Reports are generated from the Analytics module and can be exported in PDF, Excel, or CSV format. The following table summarizes the key report types available to each role, along with their primary purpose and typical usage frequency. Understanding which reports are available to each role helps organizations ensure that decision-makers have access to the right data at the right time.

Role	Report Types	Purpose
Tenant Admin	System health, user activity, audit log	Platform governance and compliance
Ops Manager	Processing throughput, inventory, shipment status	Operational efficiency monitoring
Field Officer	Farmer enrollment, harvest yields, farm health	Agricultural performance tracking
QC Inspector	Inspection results, cupping trends, defect analysis	Quality assurance and grading
Trader	Market prices, RFQ pipeline, contract status	Commercial performance analysis
Finance Manager	Revenue, costs, payment reconciliation, margins	Financial control and planning
Buyer	Shipment tracking, quality verification, spend analysis	Procurement optimization
Viewer	Executive summary, KPI dashboard, compliance status	Strategic oversight and governance

Table 5.2: Report Types by Role

5.3 Dashboard KPIs

Each entity type has a tailored Dashboard with KPIs relevant to its operational context. These KPIs provide at-a-glance performance indicators that help users quickly assess the health of their operations. The Dashboard is the first module users see after login, and its KPI cards are designed to surface the most actionable metrics. The following table lists the primary KPIs displayed on each entity type Dashboard, organized by the operational domain they represent.

Entity Type	Primary KPIs
Producer	Total farmers, total farm area, harvest volume, avg yield/ha, EUDR compliance %
Aggregator	Cherries received, lots processed, avg cupping score, inventory value, pending shipments
Exporter	Active contracts, shipments in transit, revenue, EUDR DDS count, on-time delivery %
Importer	RFQs submitted, shipments expected, customs clearance time, landed cost, quality match %
Buyer	Orders placed, shipments tracked, avg quality score, total spend, traceability coverage %
Certification Body	Assessments completed, certificates issued, avg compliance score, renewal rate
Laboratory	Samples tested, avg turnaround time, specialty grade %, QC report count, revenue

Table 5.3: Dashboard KPIs by Entity Type

5.4 Navigation Maps

Navigation within the Terra Brew platform is driven by the sidebar menu, which dynamically adjusts based on the user role and entity type. Each user sees only the modules they have access to, organized into the seven functional groups. The sidebar is the primary navigation mechanism, with each module represented by an icon and label. Clicking a module opens its primary view, which typically contains a data table with search and filter capabilities, along with action buttons for creating new records.

The sidebar groups and their modules are: **Overview** (Dashboard, Analytics), **Farm Operations** (Farmers, Farm Lands, Cultivations, Nurseries, Land Prep, Crop Monitor, Fertilizer, Pest Disease, Harvest Trace), **Processing and Quality** (Procurement, Processing, Coffee Inspection, QC Verify), **Compliance and Certification** (EUDR, Cert Assess, Deforestation), **Trade and Logistics** (Marketplace, RFQ, Inspections, Product Monitor, Smart Contracts, Trading Desk, Shipments, Logistics, Export Docs, Buyers), **Finance and Admin** (Billing, Users), and **System and Integrations** (IoT Sensors, Blockchain, API Settings). For a Field Officer at a Producer, the sidebar shows Farm Operations as the dominant group. For a Trader at an Exporter, the Trade and Logistics group dominates. For a Tenant Administrator, all groups and modules within the entity type scope are visible. The breadcrumb trail at the top of each page shows the current navigation path, allowing users to quickly return to parent views.

Chapter 6: EUDR Compliance Workflow

6.1 EUDR Data Requirements

The European Union Deforestation Regulation (EUDR) mandates that all operators placing coffee on the EU market must demonstrate that the product was not produced on land deforested after December 31, 2020. Compliance requires the submission of a Due Diligence Statement (DDS) that includes specific mandatory data elements. The Terra Brew platform systematically collects, validates, and packages these data elements to support EUDR compliance. The mandatory data elements include: the geolocation of all plots of land where the coffee was produced (as GPS polygons, not just point coordinates), the country and region of production, the name and address of the operator, evidence that the product is deforestation-free (satellite imagery analysis), and the commodity, species, and production timeframe.

Additional supporting data includes the chain of custody records showing the path of the coffee from farm to EU border, certification documentation (organic, fair trade, etc.), and risk assessment results. The platform ensures that all mandatory fields are collected at the appropriate data entry points and validated for completeness and accuracy before a DDS can be generated. For example, farm land records must include GPS polygon boundaries (not just center points), and the deforestation assessment must return a low-risk or no-deforestation finding. The EUDR module enforces these requirements through mandatory field validation and pre-submission completeness checks. If any required data is missing, the system prevents DDS generation and provides a clear list of missing elements that must be addressed.

Data Element	Source Module	Responsible Role	Validation Rule
Farm geolocation (polygon)	Farm Lands	Field Officer	Must be GPS polygon, min 3 vertices
Production country/region	Farm Lands	Field Officer	Auto-derived from GPS coordinates
Operator name and address	Tenant config	Tenant Admin	Non-empty, valid format
Deforestation-free evidence	Deforestation	Tenant Admin	Satellite assessment: low risk
Commodity and species	Cultivations	Field Officer	Must be Coffea Arabica/Canephora
Production timeframe	Harvest Trace	Field Officer	Date within harvest season
Chain of custody	Trace Journey	System-generated	Complete, unbroken chain
Certification documentation	Cert Assess	QC Inspector	Valid certificate within period

Table 6.1: EUDR Mandatory Data Elements

6.2 Deforestation Assessment Flow

The deforestation assessment is a multi-step process that combines satellite imagery analysis with farm-level data verification to determine whether a given plot of land has been deforested since the EUDR cutoff date of December 31, 2020. The assessment begins when a farm land record with GPS polygon boundaries is created or updated in the platform. The system automatically submits the polygon coordinates to the integrated satellite monitoring service, which analyzes historical and current satellite imagery to detect land cover changes within the defined boundary.

The satellite analysis produces a risk score on a scale of 0 to 100, where lower scores indicate lower deforestation risk. Scores below 30 are classified as low risk, scores between 30 and 70 as medium risk, and scores above 70 as high risk. Low-risk assessments automatically pass and can be included in the DDS. Medium-risk assessments require additional verification, such as ground-truth photographs or third-party audit confirmation. High-risk assessments block DDS generation and trigger a remediation workflow that requires the Field Officer to visit the farm, document the current land use, and submit evidence that the land was not deforested. The Deforestation module provides a visual interface showing the farm polygon overlaid on satellite imagery, with color-coded risk indicators. For farms that pass the assessment, the system generates a deforestation-free certification record that is linked to the farm land record and automatically included in the DDS when the farm contributes to an EU-bound shipment.

6.3 Cross-Border Compliance

EUDR compliance requires a seamless handoff of data between the Exporter and the Importer. The Exporter, as the operator placing coffee on the EU market, is legally responsible for submitting the DDS to the EU Information System. However, the data underpinning the DDS originates from multiple upstream entities, including Producers (farm geolocation), Aggregators (lot composition and traceability), and Certification Bodies (compliance verification). The Terra Brew platform facilitates this cross-border data handoff through a structured DDS generation workflow.

When an Exporter prepares a shipment for an EU destination, the EUDR module aggregates all required data from the contributing entities: farm polygons from the Producer, lot composition from the Aggregator, quality reports from the Laboratory, and certification records from the Certification Body. The module validates that all mandatory fields are present and that all deforestation assessments return low-risk or verified-medium-risk status. Once validated, the DDS is generated as a structured document that includes all mandatory data elements, supporting evidence, and digital signatures. The DDS is then submitted to the EU Information System on behalf of the Exporter. A copy of the DDS is shared with the Importer through the platform, enabling the Importer to verify compliance before accepting the shipment. This cross-border data handoff is recorded on the blockchain, providing an immutable audit trail that can be used to demonstrate compliance to regulatory authorities. The platform also supports periodic compliance audits, where the Certification Body can review the DDS records and verify that the underlying data is accurate.

and complete.

Chapter 7: Blockchain and Traceability

7.1 Hash Chain Architecture

The Terra Brew platform implements a hash chain architecture to ensure data immutability across the supply chain. Unlike traditional blockchain systems that require mining and consensus mechanisms, the platform uses a lightweight hash chain that links each data record to its predecessor through a cryptographic hash. When a new record is created (e.g., a harvest entry, a processing batch, a quality report), the system computes a SHA-256 hash of the record data concatenated with the hash of the previous record in the chain. This creates a tamper-evident sequence where any modification to a historical record would invalidate all subsequent hashes, making data manipulation immediately detectable.

The hash chain is stored in the Blockchain module, which provides a visual representation of the chain and tools for verifying data integrity. Users can select any record in the chain and verify that its hash is consistent with the recorded values, confirming that the data has not been altered since it was originally entered. The hash chain spans across entity boundaries, with cross-entity data handoffs creating additional chain links. For example, when an Aggregator creates a lot that includes coffee from multiple Producer harvests, the lot record includes the hashes of each contributing harvest record. This creates a Merkle-like structure that enables efficient verification of specific data points without requiring access to the entire chain. The Blockchain module also provides an API for third-party verification, allowing external auditors or regulatory bodies to verify the integrity of the traceability data without direct platform access.

7.2 QR Code Verification

QR code verification is the consumer-facing component of the Terra Brew traceability system. Each lot that passes through the platform is assigned a unique QR code that can be printed on packaging, labels, or documentation. When a consumer or supply chain partner scans the QR code using a smartphone, they are directed to a publicly accessible traceability page that presents the complete journey of the coffee from farm to cup. The traceability page is designed to be visually engaging and easy to understand, using a timeline layout that shows each stage of the coffee journey with key data points.

The QR code verification page displays the following information: the farm of origin (country, region, farm name, altitude), the farmer who grew the coffee, the harvest date and method, the processing method and parameters, the quality grade and cupping score (with attribute breakdown), the certification status (organic, fair trade, etc.), the export and shipment dates, and the EUDR compliance status. The page also includes a verification badge that confirms the data has been validated against the blockchain hash chain, providing cryptographic proof of data integrity. This QR code verification system transforms the technical infrastructure of the platform into a

consumer-facing trust signal, enabling brands to differentiate their products through verifiable transparency. The system also supports dynamic QR codes that can be updated with additional traceability data as the coffee moves through the supply chain, ensuring that the consumer always sees the most current information.

7.3 Trace Journey Module

The Trace Journey module is the internal counterpart to the consumer-facing QR verification system, providing platform users with a comprehensive, interactive view of the complete traceability chain for any lot, shipment, or batch. The module is accessible to all entity types (with View access for Buyers and Certification Bodies) and serves as the primary tool for supply chain transparency and due diligence. The Trace Journey presents a step-by-step visual timeline of the coffee journey, with each step showing the entity involved, the data recorded, the timestamp, and the blockchain hash verification status.

Users can navigate the Trace Journey in two directions: backward tracing (from a finished product to its origin farms) and forward tracing (from a specific farm to all products that contain its coffee). Backward tracing is used by Buyers to verify provenance, by Certification Bodies to audit compliance, and by regulatory authorities to investigate food safety incidents. Forward tracing is used by Producers to track where their coffee ends up, by Exporters to manage recalls, and by Aggregators to understand market distribution. The Trace Journey module also supports batch-level tracking, where users can follow a specific batch of coffee through each processing and logistics step, seeing how it is combined with other batches, split into sub-lots, or regraded based on quality results. Each step in the journey is linked to its blockchain hash, providing instant verification of data integrity directly within the trace view.

Chapter 8: Testing and Validation Checklist

8.1 Per-Role Login Credentials

The following table provides the complete list of seed user accounts provisioned in the Terra Brew platform for testing and validation purposes. Each account is pre-configured with a specific role and entity type, and all accounts share the same default password: **Admin@2024**. These accounts should be used to validate that each role sees the correct modules, has the appropriate data access, and can perform the expected actions. After initial testing, it is recommended to change all default passwords and disable any accounts that are not needed for ongoing operations.

Email	Role	Entity Type	Tenant
admin@metrang-coffee.terrabrew.com	tenant_admin	Producer	Metrang Coffee
field_officer@metrang-coffee.terrabrew.com	field_officer	Producer	Metrang Coffee

viewer@metrang-coffee.terrabrew.com	viewer	Producer	Metrang Coffee
admin@cooxupe.terrabrew.com	tenant_admin	Aggregator	Cooxupe
operations@cooxupe.terrabrew.com	operations_manager	Aggregator	Cooxupe
trader@cooxupe.terrabrew.com	trader	Aggregator	Cooxupe
admin@yirgacheffe-union.terrabrew.com	tenant_admin	Producer	Yirgacheffe Union
field_officer@yirgacheffe-union.terrabrew.com	field_officer	Producer	Yirgacheffe Union
admin@othaya-cooperative.terrabrew.com	tenant_admin	Producer	Othaya Cooperative
field_officer@othaya-cooperative.terrabrew.com	field_officer	Producer	Othaya Cooperative
admin@euro-coffee-imports.terrabrew.com	tenant_admin	Exporter	Euro Coffee Imports
trader@euro-coffee-imports.terrabrew.com	trader	Exporter	Euro Coffee Imports
operations@euro-coffee-imports.terrabrew.com	operations_manager	Exporter	Euro Coffee Imports
buyer@euro-coffee-imports.terrabrew.com	buyer	Exporter	Euro Coffee Imports
buyer2@euro-coffee-imports.terrabrew.com	buyer	Exporter	Euro Coffee Imports
admin@sgs-inspection.terrabrew.com	tenant_admin	Cert Body	SGS Inspection
inspector@sgs-inspection.terrabrew.com	quality_controller	Cert Body	SGS Inspection

Table 8.1: Seed User Credentials (Password: Admin@2024)

8.2 First-Login Navigation Guide

After logging in as each role, the following checks should be performed to verify that the platform is correctly configured and that the user experience matches the expected workflow. This guide provides a systematic approach to validating each role, ensuring that no critical functionality is overlooked during the testing phase. For each role, log in using the credentials from the previous table, then follow the verification steps listed below.

Role	First Page	Key Verifications
tenant_admin	Dashboard	All modules visible, Users module accessible, Billing tab present, API Settings functional
operations_manager	Dashboard	Procurement, Processing, Shipments visible; no Users or API Settings access

field_officer	Dashboard	Farm Operations group fully visible; no Trade/Logistics modules; mobile layout works
quality_controller	Dashboard	Coffee Insp, QC Verify, Inspections visible; no Marketplace or Trading access
trader	Dashboard	Marketplace, RFQ, Smart Contracts, Trading Desk visible; no Farm Operations access
finance_manager	Dashboard	Billing module visible with financial data; no Trade module write access
buyer	Dashboard	Marketplace, RFQ, Shipments (view), Trace Journey visible; no Processing or Export Docs
viewer	Dashboard	Dashboard and Analytics only; no create/edit buttons; all data read-only

Table 8.2: First-Login Navigation Verification Checklist

8.3 Critical Path Testing

Critical path testing validates the end-to-end functionality of the Terra Brew platform by executing the primary workflows that must work correctly for the system to be considered production-ready. These test scenarios cover the most important data flows and user interactions, ensuring that data entered at one stage of the supply chain is correctly propagated through subsequent stages and that cross-entity data handoffs function as expected. Each test scenario should be executed with the specified roles and the results verified against the expected outcomes.

Scenario	Roles Involved	Steps	Expected Outcome
Farmer Enrollment	Field Officer	1. Log in 2. Navigate to Farmers 3. Add farmer 4. Map farm land	Farmer profile created with GPS polygon
Harvest and Procurement	Field Officer, Ops Mgr	1. Record harvest 2. Log procurement at Aggregator	Harvest data linked to procurement record
Processing and QC	Ops Mgr, QC Inspector	1. Create processing batch 2. Submit sample to lab 3. Record cupping scores	Lot created with quality grade from lab results
Marketplace to Contract	Trader, Buyer	1. List lot 2. Submit RFQ 3. Negotiate 4. Sign smart contract	Contract recorded on blockchain, both parties notified
EUDR DDS Generation	Field Officer, Tenant Admin	1. Map farm land 2. Run deforestation assessment 3. Generate DDS	DDS created with all mandatory fields and low-risk assessment
Shipment Tracking	Trader, Buyer	1. Create shipment 2. Update tracking 3. Confirm delivery	Buyer receives real-time tracking updates and delivery confirmation
QR Code Verification	Buyer, Consumer	1. Generate QR code 2. Scan with phone 3. View traceability page	Complete farm-to-cup journey displayed with hash verification

Cross-Tenant Data Flow	Field Officer, Trader	1. Producer records harvest 2. Aggregator creates lot 3. Exporter lists on marketplace	Data flows correctly across tenant boundaries with hash chain intact
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Table 8.3: Critical Path Test Scenarios

Each test scenario should be executed multiple times with different data variations to ensure robustness. Edge cases to test include: harvest quantities at the boundary of acceptable ranges, cupping scores at grade transition points (e.g., 79.9 vs 80.0 for Specialty), RFQ negotiations with multiple counter-offers, and EUDR assessments for farms with medium or high risk scores. All test results should be documented with screenshots and logged in the platform test management system. Any failures should be reported with the exact steps to reproduce, the expected result, and the actual result, enabling the engineering team to diagnose and resolve issues efficiently. The critical path testing phase should be completed before any user acceptance testing (UAT) or production deployment.